Bachelor of Science (Hons) in Software Engineering

Bachelor of Science (Hons) in Software Engineering Year 01 Semester 01 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 1									
	Semester 1									
SE1101	Fundamentals of Computer Systems	C	15	30	55	100	2		30	70
SE1102	Fundamentals of Programming	C	30	30	90	150	3		40	60
SE1103	Fundamentals of Mathematics	C	30		70	100	2		30	70
SE1104	Computer Architecture and Organization	C	30		70	100	2		30	70
SE1105	Database Management Systems	C	30	30	90	150	3		30	70
SE1106	English	C	30		70	100		2	50	50
SE1107	Leadership	C	30		70	100	2		50	50
	Total Academic Credits for Semester I						**14**	**2**		

Bachelor of Science (Hons) in Software Engineering Year 01 Semester 02 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 1									
	Semester 2									
SE1208	Operating Systems	C	30	30	90	150	3		30	70
SE1209	Object Oriented Programming	C	30	30	90	150	3		30	70
SE1210	Data Communication and Networks	C	30	30	90	150	3		40	60
SE1211	Software Engineering	C	30		70	100	2		30	70
SE1212	Probability and Statistics	C	30		70	100	2		30	70
SE1213	Group Project in Programming	C	30	30	90	150	3		40	60
SE1214	Scientific Communication	C	30		70	100	2		50	50
	Total Academic Credits for Semester II						**18**	**0**		

Bachelor of Science (Hons) in Software Engineering Year 02 Semester 03 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 2									
	Semester 3									
SE2101	Object Oriented Analysis and Design	C	30		70	100	2		30	70
SE2102	Data Structures and Algorithms	C	30	30	90	150	3		30	70
SE2103	Graphics and Visual Computing	C	30	30	90	150	3		30	70
SE2104	Web Technologies	C	30	30	90	150	3		50	50
SE2105	Advanced Databases	C	30	30	90	150	3		50	50
SE2106	Advanced Mathematics	C	30		70	100	2		30	70
	Total Academic Credits for Semester III						**16**	**0**		

Bachelor of Science (Hons) in Software Engineering Year 02 Semester 04 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 2									
	Semester 4									
SE2207	Introduction to Artificial Intelligence	C	30		70	100	2		30	70
SE2208	Human Computer Interaction	C	30		70	100	2		30	70
SE2209	Software Architecture and Design Patterns	C	30		70	100	2		30	70
SE2210	Service Oriented Web Applications	C	30		70	100	2		30	70
SE2211	Deductive Reasoning and Logic Programing	C	30		70	100	2		30	70
SE2212	Statistical Distribution and Inferences	C	30		70	100	2		30	70
SE2213	Software Project Management	C	30		70	100	2		40	60
SE2214	Software Quality Assurance	C	30	30	90	150	3		50	50
	Total Academic Credits for Semester IV						**17**	**0**		

Bachelor of Science (Hons) in Software Engineering Year 03 Semester 05 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 3									
	Semester 5									
SE3101	Software Requirement Development	C	30		70	100	2		30	70
SE3102	Machine Learning	C	45		105	150	3		30	70
SE3103	Professional Practices in Computing	C	30		70	100	2		30	70
SE3104	Research Methodology	C	30		70	100	2		30	100
SE3105	Computer and Network Security	C	30		70	100	2		30	70
SE3106	Software Evolution	C	45		105	150	3		30	70
SE3107	Current Topics in Software Engineering	C	30		70	100	2		30	70
SE3108	Bioinformatics*	E	30		70	100	2		30	70
SE3109	Data Mining and Warehousing*	E	30		70	100	2		30	70
SE3110	Wireless and Mobile Computing*	E	15	30	55	100	2		30	70
SE3111	Linear Programming*	E	30		70	100	2		30	70
SE3112	Modeling and Simulation*	E	30		70	100	2		30	70
SE3113	Numerical Methods*	E	30		70	100	2		30	70
SE3114	Systems Programming*	E	30		70	100	2		30	70
	Total Academic Credits for Semester V						**18**	**0**		

* One Electives should be selected from the list

Bachelor of Science (Hons) in Software Engineering Year 03 Semester 06 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 3									
	Semester 6									
SE3220	Industrial Training	C			600			6		100
	Total Academic Credits for Semester VI						**0**	**6**		

Bachelor of Science (Hons) in Software Engineering Year 04 Semester 07 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 4									
	Semester 7									
SE4101	Software Process Improvement	C	45		105	150	3		30	70
SE4102	Software Verification and Validation	C	45		105	150	3		30	70
SE4103	Big Data Analytics	C	15	30	55	100	2		30	70
SE4104	Embedded Systems Development	C	15	30	55	100	2		30	70
SE4105	Formal Methods in Software Development	C	30		70	100	2		30	70
SE4106	Enterprise Modelling Ontology	C	30		70	100	2		30	70
SE4107	Robotics	C	30	30	90	150	3		30	70
SE4108	Research Project**	C			800	800	(8)		20	80
SE4109	Digital Forensics*	E	30		70	100	2		30	70
SE4110	Fuzzy Logic*	E	30		70	100	2		30	70
SE4111	Multi Agent Systems*	E	30		70	100	2		30	70
SE4112	Secure Programming*	E	30		70	100	2		30	70
SE4113	Computational Linguistics*	E	30		70	100	2		30	70
SE4114	Crypto Currency and Blockchain Technologies*	E	30		70	100	2		30	70
SE4115	Entrepreneurship	C	30		70	100		2	50	50
	Total Academic Credits for Semester VII						**19**	**2**		

* One Electives should be selected from the list

** The project is initiated in Semester 7 with the proposal defense followed by the project report interim submission which provides an important milestone at the end of the Semester 7, giving staff and students the opportunity to assess progress. At the end of Semester 8 the student will be required submit the final report and to attend and pass a viva voice examination and the 8 Credit is added to semester 8.

Bachelor of Science (Hons) in Software Engineering Year 04 Semester 08 Curriculum

Module Code	Module Name	Category (C/E/O)	DCH		Independent Learning Hours	NLH	Academic Credits		Evaluation	
			Lect.	Practs.			GPA	NGPA	CA %	FA %
	Year 4									
	Semester 8									
SE4216	Theory of Compilers	C	30		70	100	2		30	70
SE4217	Software Maintenance and Re-Engineering	C	45		105	150	3		30	70
SE4218	Engineering Economics	C	30		70	100	2		30	70
SE4219	Cloud Computing and IOT	C	30		70	100	2		30	70
SE4220	Deep Learning Architectures	C	30		70	100	2		30	70
SE4221	Natural Language Processing*	E	30		70	100	2		30	70
SE4222	High Performance Computing*	E	30		70	100	2		30	70
SE4223	Emerging Trends in Computing*	E	30		70	100	2		30	70
SE4224	Philosophy of Science*	E	30		70	100	2		40	60
SE4225	Game Development*	E	15	30	55	100	2		40	60
	Research Project (Started in 7th Semester)						8			
	Total Academic Credits for Semester VII						**21**	**0**		

* One Electives should be selected from the list

NOT: I added the credits to last semester.

Total Credits for the Degree Programme

	BSc (Hons) in Software Engineering	
	GPA Credits	**NGPA Credits**
Semester 1	14	02
Semester 2	18	00
Semester 3	16	00
Semester 4	17	00
Semester 5	18	00
Semester 6	00	06
Semester 7	19	02
Semester 8	21	00
Total	**123**	**10**

Semester	**1**		
Course Code:	SE1101		
Course Name:	Fundamentals of Computer Systems		
Credit Value:	02 GPA		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	15	30	55

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to,

LO1	Outline basic knowledge and understanding of the construction of computer systems and data storage
LO2	Outline working knowledge of a low level and high-level programming of hardware devices
LO3	Discuss the impact of computer hardware over its lifespan

Course Content:

- Topic 01: General Computer Organization
- Topic 02: Number Systems and Data Representation
- Topic 03: Boolean Algebra and Digital Logic
- Topic 04: CPU Organization and Instruction Set Architecture
- Topic 05: Volatile and Non-volatile Storage
- Topic 06: System Software and Utilities
- Topic 07: Operating Systems
- Topic 08: BIOS settings
- Topic 09: Evolution of Computers
- Topic 10: Computer Classification and the evolution of computers
- Topic 11: Introduction to Computer Networks
- Topic 12: System Maintenance and Troubleshooting

Practical:

- Hands on experience on Computer Hardware
- Setup basic computer system
- BIOS configurations
- Install Operating System and Productive Software applications
- Network Configurations

- Basic Troubleshooting

Teaching /Learning Methods:

This course is taught primarily face to face with active practical work interspersed with short examples and discussions. Students are expected to work on practical and review course material between face-to-face sessions.

Assessment Strategy:

Continuous Assessment 30%			Final Assessment 70%		
Quizzes (%)	Mid-term (%)	Assignment (%)	Theory (%)	Practical (%)	Other(%)
20	30	50	60	40	-

Recommended Reading:

- Lobur, J.: The essentials of computer organization and architecture. 4th ed. Jones and Bartlett Publishers, USA (2004).
- Mueller, S.: Upgrading and repairing PCs. 22nd ed. Pearson Education Ltd, USA (2015).
- Norton, P.: Introduction to Computers. 6th ed. McGraw Hill Education, New Delhi (2005).

Semester	**1**		
Course Code:	SE1102		
Course Name:	Fundamentals of Programming		
Credit Value:	03 GPA		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	30	30	90

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to,

LO1	Identify key constructs of a procedural and object-oriented programming language
LO2	Describe simple algorithmic graphics mechanism
LO3	Relate general problem-solving ideas to specific programming tasks
LO4	Create workable computer programs for stated problems or creative production.

Course Content:

- Topic 01: Computer Programming Fundamentals
- Topic 02: Data representations in Computer
- Topic 03: Graphical representation of Algorithms
- Topic 04: Generations of Computer Software
- Topic 05: Introduction to Algorithms and Programming Concepts
- Topic 06: Basics of C programming
- Topic 07: Input and Output commands
- Topic 08: Control Statements
- Topic 09: Arrays and Strings
- Topic 10: Functions
- Topic 11: Pointers in C
- Topic 12: User-Defined Data Types and Variables
- Topic 14: Files in C
- Topic 15: Advanced C

Practical:
This course module lab practical are covered by C Language. Students develop the skills of writing C programming and solving problems and the concepts like looping, array, functions, pointers, file, structure.

Teaching /Learning Methods:
This course is taught primarily face to face with active practical work interspersed with short examples and discussions. Students are expected to work on practical and review course material between face-to-face sessions.

Assessment Strategy:					
Continuous Assessment 40%			Final Assessment 60%		
Quizzes (%) 20	Mid-term (%) 30	Assignments (%) 50	Theory (%) 60	Practical (%) 40	Other (%)
Recommended Reading: • Pradip, D., Manas G.: Computer Fundamentals and Programming in C. OUP, India (2013). • Albahari, J., Albahari, B.: C# 7.0 in a nutshell. 7th ed. CA: O'Reilly Media, Sebastopol (2018). • Balagurusamy, E.: Computing fundamentals and C programming. 2nd ed. Tata McGraw-Hill Pub. Co. Ltd, New Delhi (2017).					

Semester	**1**		
Course Code:	SE1103		
Course Name:	Fundamentals of Mathematics		
Credit Value:	02 GPA		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	30	-	70

Course Aim/Intended Learning Outcomes:

At the completion of this course student will be able to,

LO1	Describe the system of real numbers
LO2	Define the algebraic expression and arithmetic operations
LO3	Identify prime numbers and its applications
LO4	Compute expressions with indices and logarithmic expressions
LO5	Discuss set properties and set laws
LO6	Interpret the functions
LO7	Construct propositions and evaluate truth values
LO8	Compute Boolean expression
LO9	Calculate the number of elements in certain mathematically defined sets

Course Content:

- Topic 01 : Introduction to Mathematics, Binomial expansions, Infinity
- Topic 02 : Factorization of algebraic expression, Equations involving one variable, Simultaneous equation systems solving
- Topic 03 : Algebraic fractions, Prime numbers, Division algorithm and Modular Arithmetic
- Topic 04 : Laws of indices, Properties of surds, Properties of logarithms and applications
- Topic 05 : Sets, Types of sets, set operations, Venn-diagrams, Set laws
- Topic 06 : Functions, One-to-one and onto mappings, Inverse functions, Composition of functions
- Topic 07 : Logic operators, Compound Propositions, Truth table, Construct propositional logic system
- Topic 08 : Boolean Algebra, Boolean expressions computation, Logic Gates

- Topic 09 : Permutations and Combinations

Tutorials:

This module tutorials are covered by the mathematical examples and the exercises.

Teaching /Learning Methods:

Content is delivered using a student-centered teaching approach. Teaching learning activities are planned as lectures, tutorial sessions, information retrieval from various sources to answer questions posted on learning management system, Q & A session among students.Tutorials:This module tutorials are covered by the mathematical examples and the exercises.

Assessment Strategy:

Continuous Assessment 30%			Final Assessment 70%		
Quizzes (%) 20	Mid-term (%) 30	Assignments (%) 50	Theory (%) 100	Practical (%)	Other (%)

Recommended Reading:

- Vince, J.: Foundation mathematics for computer science. 2nd ed. Springer, (2020).
- Datta, K.: Matrix and Linear Algebra. 1st ed. K. B. Datta, London (2004.).
- Eric,L., Frank, T. L., Meyer, A. R., Mathematics for Computer Science. Samurai Media Limited, (2017).
- H.S. Hall, R.S. Knight. : Elementary Algebra for School. Metric Edition, A.I.T.B.S. Publishers & Distributors, India (2000).
- Seymour Lipshutz, Marc Lipson. : Schaum's Outline series: Discrete Mathematics. 2 ed, Tata McGraw-Hill, India (2003).
- Olympia Nicodemi . : Discrete Mathematics. CBS publishers and Distributors, India (2001).

Semester	**1**		
Course Code:	SE1104		
Course Name:	Computer Architecture and Organization		
Credit Value:	02 GPA		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	30	-	70

Course Aim/Intended Learning Outcomes:

At the completion of this course student will be able to,

LO1	Explain contemporary computer system organization.
LO2	State data representation in digital computers.
LO3	Outline the concepts of memory hierarchy, interrupt processing, and input/output mechanisms.
LO4	Explain the interfaces between software and hardware components.
LO5	Describe the design of instruction set architectures.
LO6	Describe the concept of virtual memory and how it is realized

Course Content:

- Topic 01: Introduction, History, Ahmdahl's Law.
- Topic 02: Number Systems, Computer Arithmetic
- Topic 03: Digital Logic
- Topic 04: Bus Architectures
- Topic 05: Cache Memory, Internal Memory
- Topic 06: External Memory, Input and Output Devices
- Topic 07: I/O, Operating System Support
- Topic 08: Instruction Sets, Intro to Assembly Language
- Topic 09: Addressing Modes, Processor Structure and Function
- Topic 10: CPU: RSIC vs. CISC Architecture
- Topic 11: RISC, Instruction Level Parallelism
- Topic 12: Instruction Level Parallelism, Parallel Processing
- Topic 13: Parallel Processing, Multicore Computers.
- Topic 14: GPGPU, Control Unit

Teaching /Learning Methods:

Content is delivered using a student-centred teaching approach. Teaching learning activities are planned as lectures, information retrieval from various sources to answer questions posted on learning management system, Q & A session and group presentations among students.

Assessment Strategy:

Continuous Assessment 30%			Final Assessment 70%		
Quizzes (%)	Mid-term (%)	Assignments (%)	Theory (%)	Practical (%)	Other (%)
20	30	50	100		

Recommended Reading:

- Stallings, W.: Computer organization and architecture. 10th ed. Pearson, India (2016).
- Page, D.: Practical Introduction to Computer Architecture. Springer, London (2009).

Semester	**1**		
Course Code:	SE1105		
Course Name:	Database Management Systems		
Credit Value:	03 GPA		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	30	30	90

Course Aim/Intended Learning Outcomes:
At the completion of this course student will be able to,

LO1	Identify the main stages in designing and implementing Relational Databases
LO2	Describe main concepts related to relational databases
LO3	Outline well-defined problem, design and implement an optimized relational database to solve it
LO4	Produce simple queries by matching relational algebraic fundamentals
LO5	Outline SQL constructs to write queries to manipulate data

Course Content:

- Topic 01: Introduction to Databases
- Topic 02: Conceptual Data Modelling and Database Design
- Topic 03: The Relational Data Model and SQL
- Topic 04: Database Programming Techniques
- Topic 05: Object, Object-Relational, and XML: Concepts, Models, Languages, and Standards
- Topic 06: Database Design Theory and Normalization
- Topic 07: File Structures, Hashing, Indexing, and Physical Database Design
- Topic 08: Query Processing and Optimization
- Topic 09: Transaction Processing, Concurrency Control, and Recovering
- Topic 10: Distributed Databases, NOSQL Systems, Cloud Computing, and Big Data
- Topic 11: Advanced Database Models, Systems, and Applications
- Topic 12: Additional Database Topics: Security

Practical:

This module practices are covered by SQL Query Language and suitable tool used to cover the laboratory sessions.

Teaching /Learning Methods:

Content is delivered using a student-centred teaching approach. Teaching learning activities are planned as lectures, practical sessions, tutorial sessions, information retrieval from various sources to answer questions posted on learning management system, Q & A session among students.

Assessment Strategy:

Continuous Assessment 30%			Final Assessment 70%		
Quizzes (%)	Mid-term (%)	Assignments (%)	Theory (%)	Practical (%)	Other (%)
20	30	50	60	40	

Recommended Reading:

- Elmasri, R., Navathe, S.: *Fundamentals of Database Systems*. 7th ed. Pearson (2015).
- Coronel, C., Morris, S.: Database Systems: Design, Implementation, & Management. 13th ed. Cengage Learning, (2018).

Semester	**1**		
Course Code:	SE1106		
Course Name:	English		
Credit Value:	02 NGPA		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	30	-	70

Course Aim/Intended Learning Outcomes:

At the completion of this course student will be able to,

LO1	Recall and use new academic vocabulary.
LO2	Recognize the use of grammar in speaking and writing.
LO3	Recognize inconsistencies between pronunciation and spelling.
LO4	Reproduce moderately complex sentences with basic transitions to connect ideas in spoken and written discourse.
LO5	Identify written and spoken language patterns; including sequential events, cause and effect, compare and contrast, and problem/solution narratives.

Course Content:

- Topic 01: Reading: Writing: write describing impressions, feelings and experiences and to write about familiar topics.
- Topic 02: Listening: understand familiar topics and speech on a variety of subjects such as work, leisure and the main points when listening to current affairs.
- Topic 03: Speaking: talk about familiar topics and to give explanations and reasons for opinions, past actions and future plans. Also to recount the story of a book or film and deal with most situations in an English speaking environment.
- Topic 04: Grammar: understand and apply the body of rules surrounding English grammar in everyday contexts, including the use of nouns, adjectives, verbs, prepositions, tenses, sentence structure and phrases. Academic Writing and Grammar.

<table>
<tr><td colspan="6">Teaching /Learning Methods:

Content is delivered using a student centred teaching approach. Teaching learning activities are planned as lectures,.</td></tr>
<tr><td colspan="6">Assessment Strategy:</td></tr>
<tr><td colspan="3">Continuous Assessment

50%</td><td colspan="3">Final Assessment

50%</td></tr>
<tr><td>Quizzes (%)

20</td><td>Mid-term (%)

30</td><td>Assignments (%)

50</td><td>Theory (%)

100</td><td>Practical (%)</td><td>Other (%)</td></tr>
<tr><td colspan="6">Recommended Reading:

• Wisniewska, I., Riggenbach, H., Samuda, V.: Grammar dimension 2: Form meaning, and use. 4th ed. MA: Heinle Cengage Learning, Boston (2007).
• Honigsfeld, A.: Growing Language and Literacy: Strategies for English Learners. Heinemann, (2019).</td></tr>
</table>

Semester	**1**		
Course Code:	SE1107		
Course Name:	Leadership		
Credit Value:	02 GPA		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	15	30	55

Course Aim/Intended Learning Outcomes:

At the completion of this course student will be able to,

LO1	Define the functional Qualities and approaches to Leadership.
LO2	Identify factors which influence the morale.
LO3	Recognise the necessity for inter-relationship loyalty and discipline.
LO4	Outline the principles of good man-management.
LO5	Illustrate the importance of communication in the leadership process.

Course Content:

- Topic 01: Leadership traits and qualities
- Topic 02: The nature of leadership
- Topic 03: Leadership and management
- Topic 04: Leader behavior,
- Topic 05: Style of leadership
- Topic 06: The development process.

Teaching /Learning Methods:

Content is delivered using a student centred teaching approach. Teaching learning activities are planned as lectures, practical sessions, tutorial sessions, information retrieval from various sources to answer questions posted on learning management system, Q & A session among students.

Assessment Strategy:

Continuous Assessment 30%			Final Assessment 70%		
Quizzes (%)	Mid-term (%)	Assignments (%)	Theory (%)	Practical (%)	Other (%)

20	30	50	60	40	0%

Recommended Reading:

- Sadler, P.: *Leadership Theory & Practice*. British Library, USA,(2003)
- Edwin, A. :*The Essence of Leadership: The Four Keys to Leading Successfully,(* 1999*).*
- Peter G. Northouse.: Leadership: Theory and Practice, 6th Edition 6th Edition, SAGE Publications, Inc.(2012).
- Peter G. Northouse.: Introduction to Leadership: Concepts and Practice, 5th Edition, , SAGE Publications, (2020).